

GeoScreen- Arsenic and Uranium Geochemical Screening

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1. Background

Arsenic (As) and uranium (U) occur naturally in bedrock and soils. Elevated concentrations can be environmentally relevant, depending on concentration, medium, land use, mobility, and exposure pathways.

This mini-project builds a simple screening method to identify sample points with relatively high As and U concentrations in Swedish till soils, using SGU's regional soil geochemistry data. Candidate areas are identified for further investigation, especially where As and U are both elevated at the same point. This is a first step.

Beyond this initial step, future development will add more input data, e.g., sediment, surface soil, pH, land use, and bedrock geology. The project will then move toward spatial interpolation and machine learning. The project will then move toward spatial interpolation and machine learning. The aim is to move from identifying relative geochemical highs to predicting areas of potential risk. Later stages will focus on more detailed, site-specific analysis.

2. Data and Method

Data source:

- **Product:** [markgeokemi-regional](#), regional provtagning (SGU); layer: moran_0063mm_hno3_icpms
- **Material:** Till (moraine), fine fraction <0.063 mm
- **Method:** HNO₃ leaching / ICP-MS
- **CRS:** SWEREF99 TM (EPSG:3006)

Cleaning steps:

1. Dropped rows missing sample ID, coordinates, or geometry.
2. Confirmed CRS as EPSG:3006.
3. Filtered to rows with valid, positive values for As, Fe, Ca, Al, and U. Excluded SGU's "0 = not analyzed" placeholder and below-detection-limit values (stored as negative numbers, per SGU documentation).

Output

27 981 analysis-ready sample points (original dataset: 28 471 points). The computed percentile-based thresholds: 75th percentile (P75) → **elevated** and 95th percentile (P95) → **high or hotspots**.

Element	P75	P95
As	3.90	13.00
U	2.40	4.60

Table 1. Percentile-based thresholds for arsenic (As) and uranium (U).

Regulatory guideline values (Naturvårdsverket): for As, generic guideline values for contaminated land are 10 mg/kg TS (KM, sensitive land use, e.g. homes, schools, and gardens) and 25 mg/kg TS (MKM, less sensitive land use, e.g. offices and industry). The computed 90 percentile for As (8.4 ppm) of the SGU’s regional data falls slightly below the KM value but the P95 (13.00 ppm) sits between KM and MKM.

Supplementary geological layer, alum shale (Alunskiffer):

Product: [Berggrund 1:50 000–1:250 000](#) (SGU bedrock geology). The GeoPackage was downloaded from SGU: [berggrund50k-250k.zip](#). Alum shale polygons (n = 123) were extracted, reprojected to EPSG:3006, and overlaid with the Priority 1 and Priority 2 results (see Figure 5).

3. Results

As hotspots (top 5%, red) among all cleaned sample points (Fig. 1).

U hotspots (top 5%, blue) among all cleaned sample points (Fig. 2).

Priority 1 (red) and Priority 2 (orange) points overlaid on the full sample set (Fig. 3).

Points classified against Sweden’s legal contaminated-land guideline values (Naturvårdsverket) (Fig .4).

Priority 1 and Priority 2 points, with mapped alum shale bedrock polygons (black, n=123) (Fig .5).

Point counts per class (relative percentile screening, n=27 981):

Class	Count	% of total
As hotspot (top 5%)	1 394	4.98%
U hotspot (top 5%)	1 370	4.90%
Priority 1 (both hotspots)	137	0.49%
Priority 2 (either hotspots)	2 490	8.90%
Background	25 354	90.61%

Table 2: Point counts per class (relative percentile screening, n=27 981)

Class	Count	% of total
Below KM (<10 ppm)	25 862	92.42%
KM-MKM (10-25 ppm)	1 752	6.26%
Above MKM (>25 ppm)	367	1.31%

Table 3: Point counts per class (As vs. Naturvårdsverket guideline values, n=27 981)

4. Interpretation

The screening shows clear geographic clustering of high As and U values, mainly in northern and parts of central Sweden (Fig. 1 & 2). This method identifies points that

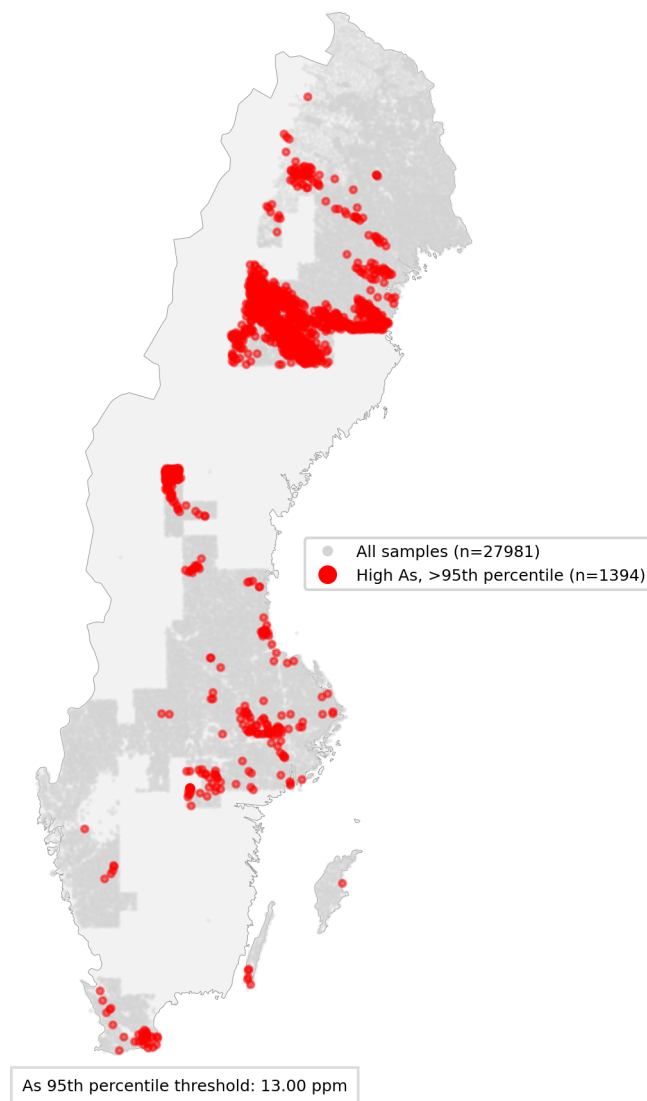


Figure 1: Arsenic screening map

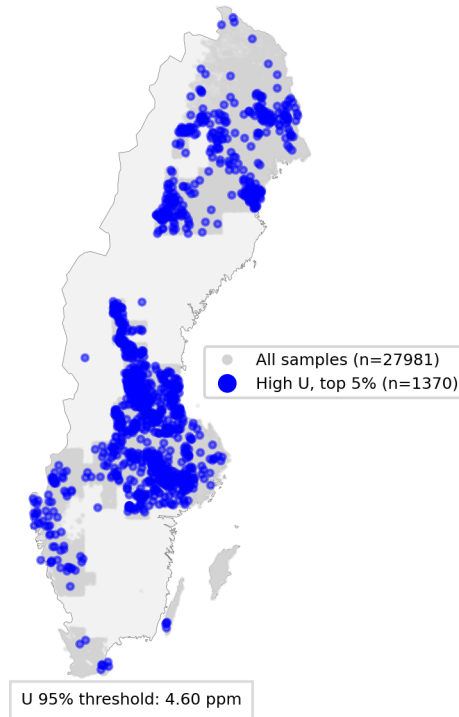


Figure 2: Uranium screening map

are high relative to this dataset (Table 2). It does not confirm regulatory exceedance on its own.

Priority 1 points (both As and U high) are the strongest signal, since two independent measurements agree (Fig. 3). Priority 2 points (only one element high) are weaker and need more context before being treated as risk areas.

Guideline comparison: 7.6% of points (2,119 of 27,981) exceed the Naturvårdsverket KM guideline value (10 ppm As) (Table 3). This is more than the 5% implied by the relative screening threshold, because KM sits below this dataset’s own 95th percentile (13.00 ppm). Guideline-based screening identifies more points than percentile-based screening.

Alum shale overlay (Figure 5): Priority 1 points in northern and central Sweden appear near mapped alum shale polygons on the map. Alum shale is known to carry elevated uranium and sulfide-hosted trace metals, including arsenic (Falk et al., 2006; Lecomte et al., 2017). Lecomte et al. (2017) also found that alum shale in northern Sweden underwent stronger metamorphism than in the south, which mobilized uranium into new minerals there. This gives a plausible geological reason for a north-south difference, but it has not been tested against this dataset.

This spatial pattern is a hypothesis, not a confirmed finding. Only 5 of 137 Priority 1 points (3.6%) fall inside a mapped shale polygon. This number alone does not confirm or rule out a real association, since no distance-based test or random baseline has been run. Section 6 lists this as a planned next step.

Southern clusters (Skåne, Gotland) show little nearby shale in this dataset. The cause is not known.

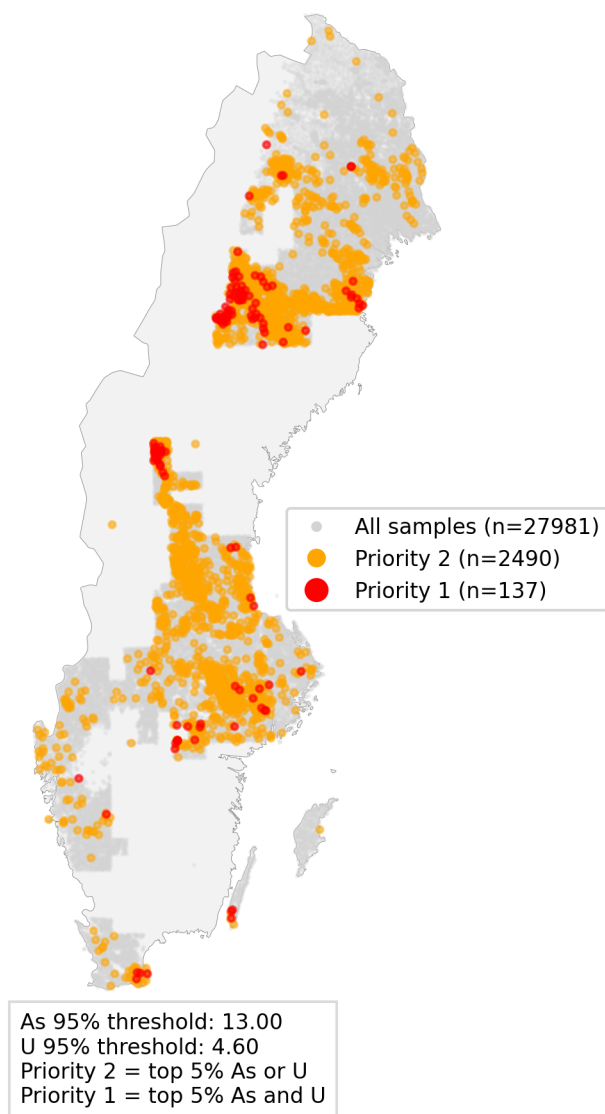


Figure 3: Combined priority map for As and U

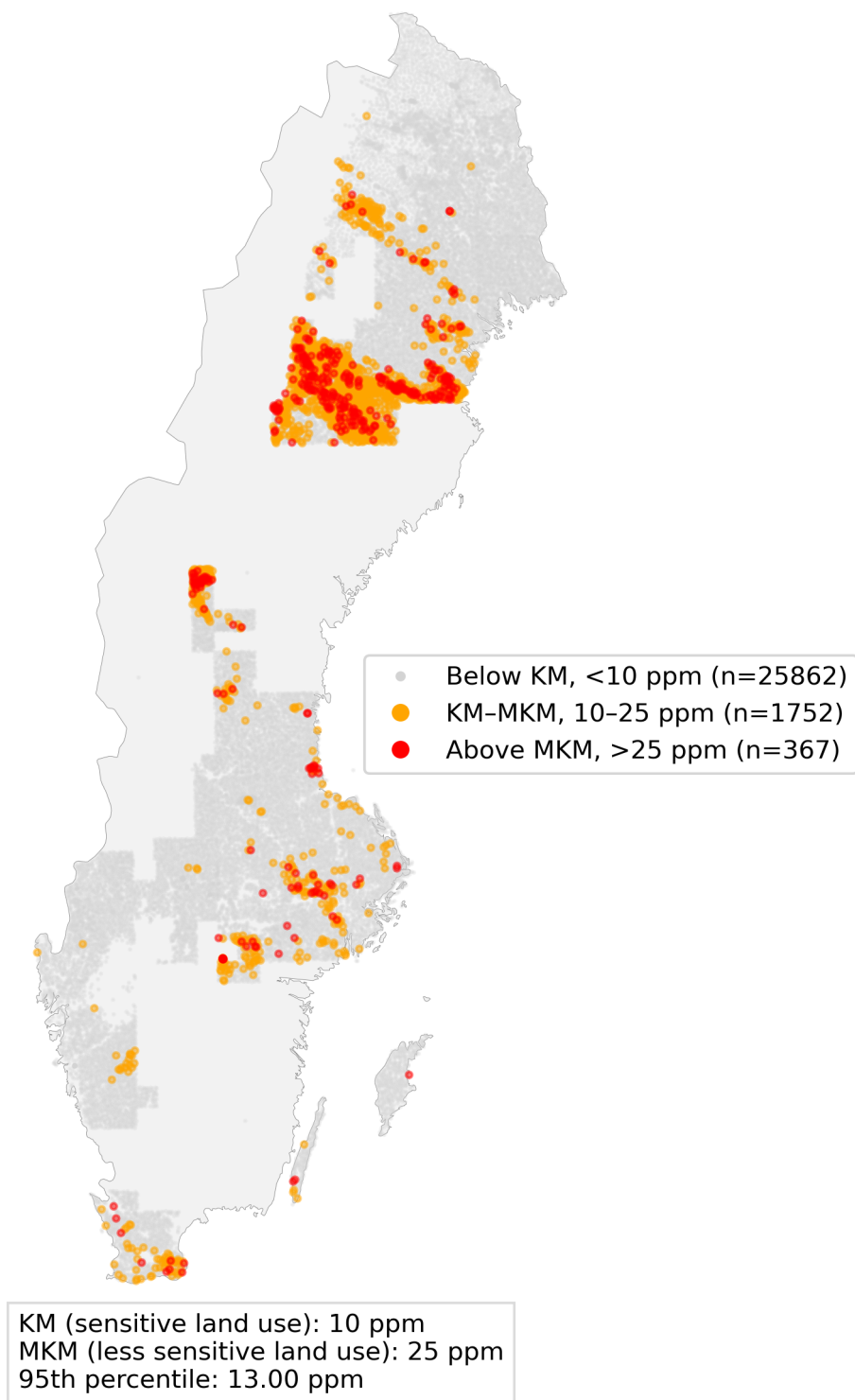


Figure 4: Arsenic vs. Naturvårdsverket guideline values (KM/MKM)

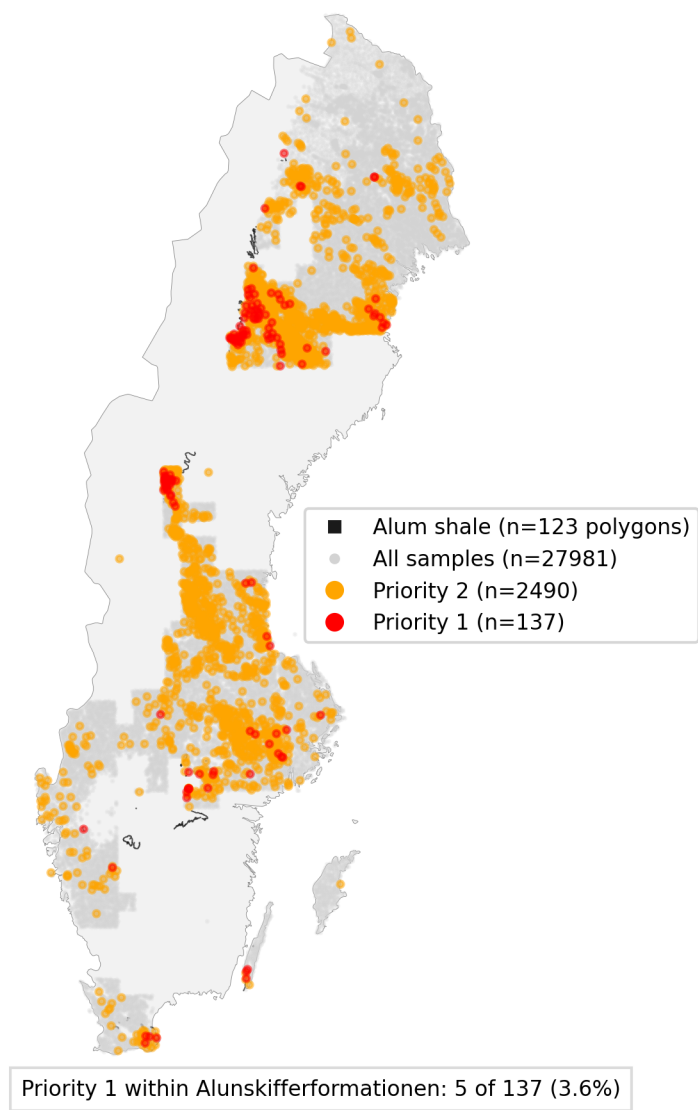


Figure 5: Combined As + U priority with alunskiffer overlay

This is a screening tool, not a risk map. It does not account for bioavailability, land use, or exposure pathways.

5. Limitations

- Classification is based on total/leachable concentrations, not bioavailability or chemical speciation.
- Relative percentiles show anomalies within this dataset, not regulatory exceedances.
- No Naturvårdsverket generic guideline value exists for uranium in soil.
- Local land use, exposure pathways, and receptor information are not included in this version.
- Groundwater conditions, pH, redox environment, and carbonate chemistry are not part of the model.
- The > 0 filter excludes “not analyzed” placeholders and below-detection-limit values, slightly narrowing the dataset.
- The alum shale overlay is untested. It does not account for glacial transport distance, ice-flow direction, or shale occurrences missing from this dataset.

6. Next Steps

1. Compare against relevant guideline values for uranium (drinking water or radiological guidance).
2. Add land use or receptor information: residential areas, agricultural land, drinking water interests, private wells.
3. Keep soil/till, sediment, surface water, and groundwater separate, since risk logic differs by medium.
4. Use Fe, Al, Ca, pH to interpret binding, mobility, and transport.
5. Test county-level thresholds instead of one national threshold, to account for natural geological background.
6. Consider log-transformation before percentile calculation, given the right-skewed nature of geochemical data.
7. Document data coverage, analytical method, sample medium, and spatial resolution in future versions.
- 8.

Run a distance-based spatial join: compute nearest-distance to shale for each point, and compare Priority 1, Priority 2, and Background classes using a statistical test (e.g. Mann-Whitney U).

7. Way Forward: Toward a Machine Learning-Based Model

The current version is a percentile-based screening method. Planned extensions:

More data:

- Additional SGU layers (`moran_2mm_hno3_icpms`, sediment, surface soil) to compare As/U across sample media.
- Additional geochemical variables: pH, S, rare earth elements.
- External layers: bedrock geology, land use, drinking water proximity.

Toward machine learning:

KNN regression will test whether Ca concentration can help predict U concentration at unsampled points. Two versions are planned: a manual $k=1$ implementation, and a `scikit-learn` version sweeping k from 1 to 70 to observe the bias-variance tradeoff.

This is exploratory work, separate from the current screening pipeline. It answers a different question (can one variable predict another) than the percentile method (which points are relatively elevated). Planned future steps:

- Spatial interpolation (kriging, IDW, or KNN) to move from points to continuous surfaces.
- Regression/classification using Fe, Ca, Al, pH as predictors of mobility.
- Train/test validation for future predictive models.
- A risk-informed prioritization model combining geochemical prediction, land use, and exposure pathways.

8. References

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